

KRYSTAL WANG

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EDUCATION

Duke University 2024 – 2028

B.S. in Biomedical Engineering | Minor in Computational Biology and Informatics | GPA: 4.0

RESEARCH EXPERIENCE

Bioengineering Summer Undergraduate Research Experience, Princeton University

Summer 2026

- Investigated the long-range communication between leader cells of migrating epithelial monolayers by stenciling then imaging tissue collisions on collagen and hydrogel plates.
- Analyzed collision data with image processing programs, segmenting cells with CellPose, tracking cell motion between frames with Fiji, then extracting data for quantitative analysis and graphical representation
- Presented findings at poster symposium

Gerecht Lab, Duke University

Jan 2026 - Present

- Engineer *in vitro* arteries by electrospinning and rolling fibrinogen into vascular grafts, then seed with endothelial cells and connect with bioreactor to simulate pulsating blood flow.
- Conduct assays and fluorescent imaging to evaluate impact of oxidative stress on blood vessels.

Arya Lab, Duke University

May 2025 – Dec 2025

- Developed a Python script that detects geometric patterns in self-assembling DNA origami; used to automate evaluation of lattice yield change after design modifications from trained neural network models.
- Followed online machine learning courses and optimized neural networks by screening through different combinations of layers.
- Organized and catalogued a structured database of 11,000+ generated data points to support ongoing analysis.
- Presented research progress at weekly lab meetings and participated in brainstorming sessions with P.I. and graduate mentor.

PROJECTS

Low-Fidelity Prosthetics, Duke eNable Club

Sep 2025 – Present

- Design low-fidelity prosthetic devices using CAD software for individuals with limb differences.
- Develop signal-processing methods to filter electrical (EMG) signals from the biceps to predict arm movement.

Salamander Sampling Deck, Duke Forest

Sep 2024 – Dec 2024

- Designed and built a new device for sampling salamander populations to support ecological monitoring in Duke Forest.
- Led a 4-person team through 5+ design iterations, integrating individual components into a functioning prototype.
- Communicated progress through weekly client meetings and authored technical memos detailing design decisions.

WORK EXPERIENCE

Student Development Representative, Duke Annual Fund

Sep 2025 – Apr 2026

- Telephoned Duke alumni across class year and department to update their profile and solicited over \$5,000 in gifts to the Duke Annual Fund in one year.

TOEFL Teaching Assistant

July 2024

- Coordinated daily with supervisors to align on teaching agendas and student learning goals.
- Provided hour-long mentoring sessions covering all TOEFL sections and graded mock exams within 24 hours.

CAMPUS & COMMUNITY INVOLVEMENT

Social VP, Duke Chinese Student Association

Sep 2025 – Present

- Plan and organize monthly social events for the Duke Chinese student community, coordinating with finance and outreach chairs on budgeting and promotion.

Co-Leader, Wild Ones Club

Mar 2025 – Present

- Coordinate and lead student trips with professors and field experts across related disciplines.

Duke Chorale

Fall 2024 - Present

Duke Swing Dance

Spring 2026 - Present

CORE SKILLS

Laboratory: Mammalian cell culture, pipetting, gel electrophoresis, cellular assays, recombinant DNA, experimental design

Computational: Python, statistical analysis, Excel, PowerPoint

Other: CAD, Laser Cutting, Literature review, scientific writing, Mandarin (fluent), Communication